

Visual P300 Oddball Paradigm

Due Date: 22nd August, 2026

Submission link: Google Form

Objective: Calculate the necessary trial counts, define precise stimulus timings, and establish confound controls for a standard visual P300 oddball experiment. Complete the parameters below.

Part 1: Trial Mathematics

You require a strict minimum of **40 clean, artifact-free Target epochs** for your final analysis to achieve a viable Signal-to-Noise Ratio. Assume a standard **15% data loss** due to ocular and muscular artifacts during preprocessing. The paradigm uses a 20% Target / 80% Standard probability ratio.

Calculate the following:

- Total Target Trials to present: _____
- Total Standard Trials to present: _____
- Total Experiment Trials: _____

Knowledge Check:

- Why is it necessary to maintain a high percentage of Standard trials (e.g., 80%) rather than a 50/50 split?

Part 2: Trial Micro-Anatomy

Define the precise millisecond durations for a single trial sequence.

Define the timings:

- Inter-Trial Interval (ITI) duration: _____ to _____ ms
- Fixation Cross duration: _____ ms
- Stimulus duration: _____ ms
- Response Window duration: _____ ms

Knowledge Check:

- Why must the ITI be jittered across a range rather than kept at a fixed, predictable duration?
- In the context of visual evoked potentials, what happens if the stimulus is left on the screen for too long (e.g., 1000 ms)?

Part 3: Session Organization

Based on your total trial count from Part 1, and assuming an average trial length of 2,700 ms, organize the experiment to manage fatigue.

Define the structure:

- Number of trials in the Practice Block: _____

- Number of recorded blocks: _____
- Trials per recorded block: _____
- Duration of mandatory rest period between blocks: _____ seconds

Knowledge Check:

- Why must the practice block trials be strictly excluded from the final data matrix?
- What specific electrophysiological changes occur in the continuous EEG if a task block runs uninterrupted for 15 minutes?

Part 4: Confound Controls

Knowledge Check:

- **Sensory Confounds:** How do you ensure the P300 amplitude difference between the Target and Standard is driven by cognitive context updating and not by low-level sensory variances (P1/N1)?
- **Motor Confounds:** The subject is instructed to press a button when they see a Target. How do you prevent the Lateralized Readiness Potential (LRP) from physically overlapping and contaminating your P300 average?